

SAURABH KUMAR

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SUMMARY

Full-Stack Developer and Computer Science undergraduate with hands-on experience building production web applications using React.js, Node.js, Express.js, FastAPI, and MongoDB/SQL. Skilled in designing scalable RESTful APIs, implementing robust authentication (OAuth 2.0, JWT), real-time communication via WebSockets, containerizing microservices with Docker on AWS EC2, and integrating AI models into user experiences.

EDUCATION

Dr. APJ Abdul Kalam Technical University

Bachelor of Technology in Computer Science and Engineering; CGPA: 7.26/10.0

Lucknow, India

Aug. 2023 – May 2027

Uttar Pradesh Board (UPMSP)

Secondary School Certificate (Class X); Percentage: 87%

Uttar Pradesh, India

May 2020

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, Python, C++, HTML5, CSS3, SQL

Frontend Development: React.js, Vite, Next.js, Tailwind CSS, Redux Toolkit, Context API, Responsive Design, SPA

Backend & APIs: Node.js, Express.js, FastAPI, RESTful APIs, WebSockets (Socket.io), JWT, OAuth 2.0, Helmet.js

Databases & ORM: MongoDB, MongoDB Atlas, MySQL, PostgreSQL, Mongoose

DevOps & Cloud: Docker, AWS EC2, Nginx, GitHub Actions (CI/CD), Vercel, Render, Git, Postman, Linux/Bash

AI Integration: LangChain, LangGraph, Groq LLM Function Calling, Google Gemini API, RAG Pipelines, Vector DBs

Core Competencies: Data Structures, Algorithms, System Design, Microservices, API Optimization

PROJECTS

HavenTo – Accommodation Booking Platform | React, Node.js, FastAPI, MongoDB, Docker, AWS EC2

Mar. 2026

- Architected full-stack property booking platform supporting property discovery, reservation lifecycle, and multi-tier host management.
- Engineered an autonomous AI travel concierge using Groq LLM with function/tool calling (property lookup, booking queries) and RAG conversation memory.
- Built a dynamic pricing microservice with REST endpoints, exposing sub-150ms real-time rate recommendations to host dashboard with one-click adjustments.
- Containerized services with Docker and deployed to AWS EC2 behind an Nginx reverse proxy with SSL; established automated GitHub Actions CI/CD for testing and deployment.
- Implemented secure JWT authentication, OTP email verification, Socket.io real-time chat, and Cloudinary media pipelines with client-side image compression.
- Links: [GitHub](#) | [Live Demo](#)

StudyMate – AI-Powered Collaborative Workspace | React, Node.js, Express, MongoDB, Gemini AI

Oct. 2025

- Engineered full-stack educational platform with AI-assisted learning, automated document summarization (PDF/DOCX), and collaborative study rooms.
- Implemented an interactive Obsidian-style Knowledge Graph using 2D force-directed physics simulation with bidirectional markdown backlinks and automatic tag clustering.
- Developed student retention and study analytics dashboard tracking weekly hours, streak consistency, and inactivity signals with predictive retention scoring.
- Integrated Google OAuth 2.0 Single Sign-On, JWT session guards, rate-limiting middleware, and Helmet.js for enterprise-grade security.
- Deployed distributed architecture on Vercel (frontend) and Render (backend) with MongoDB Atlas for high-availability data persistence.
- Links: [GitHub](#) | [Live Demo](#)

Brain Tumor Detection Portal | Python, TensorFlow, Keras, Express.js, React, OpenCV

May 2025

- Engineered full-stack medical diagnosis portal achieving 95.59% accuracy in detecting brain tumors from MRI scans using custom CNN with Transfer Learning (VGG16).
- Integrated an Express.js REST API with Python deep learning models via asynchronous child processes to stream image uploads and deliver instant inference results.
- Designed responsive hospital dashboard in React featuring secure license-based registration, Multer file uploads, and interactive confidence score visualization.
- Preprocessed and normalized 3,000+ brain MRI scan images using OpenCV and NumPy for model training, validation, and testing.
- Links: [GitHub](#)